

Yao Fan 范尧

CONTACT	<i>Phone:</i> +1 412 452 8917 <i>ORCID:</i> 0000-0002-5660-3201 <i>Personal Website:</i> https://yao-fan.github.io	<i>E-mail:</i> yao.fan@pitt.edu <i>PhilPeople:</i> yao-fan-1
AREA OF SPECIALIZATION	Social Epistemology, Philosophy of AI, Decision Theory and Game Theory, Philosophy of Statistics, Philosophy of Economics, Theories of Justice, Logic	
AREA OF COMPETENCY	Philosophy of Science, History of Modern Philosophy, Ethics, Social and Political Philosophy	
EDUCATION	University of Pittsburgh , Pittsburgh, PA, United States	
	Ph.D., Philosophy	September 2017 - July 2023
	• Dissertation: Individuals and Institutions: Essays on Social and Political Epistemic Systems (Access through Pitt D-Scholarship) • Committee Members: James Shaw (Co-Chair), Kevin Zollman (Co-Chair, CMU), Kevin Dorst (MIT), Anil Gupta	
	M.Sc., Intelligent Systems (from School of Computing and Information)	
	• Thesis Title: Personalized Decision Supports based on Theory of Mind Modeling and Explainable Reinforcement Learning (accepted by IEEE SMC 2023)	
	Australian National University , Canberra, ACT, Australia	
	Ph.D. Candidate, Philosophy (transferred to the U.S.)	February - August 2017
	B.A.(Honours), Philosophy	Feb - Dec 2016
	• <i>First Class Honours</i> • Thesis Title: “Triviality Arguments against Computational Functionalism” • Advisors: Phil Dowe, Alan Hájek, Daniel Stoljar, Koji Tanaka	
	University of Sydney , Sydney, NSW, Australia	
	B.A., Mathematics (Advanced) and Philosophy	March 2014 - December 2015
	• <i>Pass with High Distinction</i>	
EMPLOYMENT	Postdoctoral Researcher , Sun Yat-sen University	September 2025 - August 2027
	Visiting Scholar at Rutgers University Department of Philosophy from June 2026 to May 2027	
	Research Associate , Bridgewater Associates	July 2023 - August 2025
	Research on Global Macroeconomics and Apply AI to Portfolio Management	
	• <i>Bridgewater Associates is the largest hedge fund in the world, participating in global financial market based on fundamental macroeconomic research.</i>	
	Graduate Researcher , University of Pittsburgh	September 2022 - July 2023
	Research on Artificial Social Intelligence	
	• <i>Sponsored by Prof. Michael Lewis (Pitt SCI) and Prof. Katia Sycara (CMU Robotics)</i>	
	Graduate Teaching Assistant , University of Pittsburgh	September 2020 - August 2021
	Fully responsible for the teaching and grading for 3 sections, with 20 students in each section.	

- *Concepts of Human Nature: this course reads selected writings by Plato, Hobbes, Rousseau, Marx and Freud, with a focus on the historical developments of different conceptions of human nature.*
- *History of Modern Philosophy: this course reads selected writings by Descartes, Locke, Hume and Kant, with a focus on epistemology.*

Graduate Research Assistant, University of Pittsburgh **September 2018 - August 2020**

Sponsored by Prof. Robert Brandom

- *Research on Expressivist Logic and Dialogical Pragmatics.*
- *Contributed to the editorial work for **A Spirit of Trust** (Harvard University Press, 2019).*

Graduate Teaching Assistant, Australian National University **February - June 2017**

Fully responsible for the teaching and grading for 1 section of 30 students.

- *Intermediate Logic: this course covers the semantics and syntax of propositional and first-order logic, with an emphasis on proofs and Logic Programming.*

PUBLICATION

Algorithmic Monoculture and Group Decision Making: An Epistemic Braess Paradox
(draft available upon request)

- *Under Review at **Synthese***

The Inapplicability of Nash Equilibrium to Condorcet Jury Theorem or: Why I'd rather vote sincerely and so should you (draft)

- *Presented at Rutgers Philosophy (2025), Society for Exact Philosophy (2026)*
- *Under Review at **Philosophical Studies***

Regulating Strategic Track-Record Building: Institutional Remedies for Individual Expert Identification (Publisher, Preprint)

- ***Synthese**, 2026, volume 208, article 11*

Personalized Decision Supports based on Theory of Mind Modeling and Explainable Reinforcement Learning (IEEE, arXiv)

- *Li, H., **Fan**, Y., Zheng, K., Lewis, M., & Sycara, K.*
- *IEEE International Conference on Systems, Man, and Cybernetics, 2023*

Individuals and Institutions: Essays on Social and Political Epistemic Systems

- Ph.D. Dissertation, University of Pittsburgh ([Access Here](#))

SELECTED AWARDS

Forethought Fellowship, Forethought Foundation, Oxford UK

- *Research on Distributed vs Centralized Decision-Making in AI*
- *10,000 GBP*

2021 - 2023

Global Priority Institute Early Career Summer Fellowship, University of Oxford

- *Award for promising early career researchers in Philosophy and Economics* **Summer 2021**

International Postgraduate Research Scholarship, Australian National University

- *4-year merit-based scholarship awarded to 8 incoming international PhD students each year at ANU* **2017-2020 (curtailed after transfer)**

SELECTED
TALKS

Systematic Perspectives on Epistemic Deference: Relying on Others and Relying on AI

- *Institute of Philosophy, Chinese Academy of Science, Beijing, China* **May 2026**

The Epistemic Dimension of Group Decision Making

- *Philosophy Department, Peking University, Beijing, China* **May 2026**

From Marx to Rawls and Beyond: Or, How I learnt to Stop Worrying and Love “Bourgeois” Theories of Justice

- *Bridgewater Associates Internal Conference, United States* **March 2024**

Science and Commonsense: An Inferentialist Approach toward Primitivism of Colors

- *12th Arché Graduate Conference, University of St Andrews, U.K.* **February 2021**
- *Shanxi University, Taiyuan, China* **December 2019**
- *Pitt Graduate Work-in-Progress* **October 2019**

From General Logic to Transcendental Logic

- *Pittsburgh Area Kant Colloquium* **October 2020**

Infinite Gambles in Savage’s Decision Theory

- *The 24th World Congress of Philosophy* **August 2018**

Triviality Arguments against Computational Functionalism

- *Philosophy Honours Colloquium, Australian National University* **September 2016**

VISITING

Department of Philosophy, Rutgers University, NJ, United States

Sponsored by Prof. Susanna Schellenberg **June 2026 - July 2027**

Global Priorities Institute, University of Oxford, Oxford, United Kingdom

Sponsored by GPI Early Career Fellowship **June - July 2021**

Summer School in Logic and Formal Epistemology, Pittsburgh, PA, United States

Hosted by Department of Philosophy, Carnegie Mellon University **June 2018**

Nordic Logic Summer School, Stockholm, Sweden

Sponsored by Nordic Logic Summer School Participation Grant **August 2017**

ANU Logic Summer School, Canberra, Australia

Sponsored by Bob Meyer Memorial Scholarship **December 2016**

Higher School of Economics, Moscow, Russia

Visiting Student specialized in Cognitive Science & Philosophy **September - December 2015**

University of California, Davis, CA, United States

Exchange Student specialized in Mathematics & Philosophy **January - July 2015**

University of California, Los Angeles, CA, United States

Summer Courses on Political Science and Political Philosophy **June - August 2014**

PROFESSIONAL SERVICE	Reviewer	
	• <i>Synthese</i> • IEEE International Conference on Systems, Man, and Cybernetics (2023, 2026)	
	University of Pittsburgh	
	Pitt-CMU Graduate Conference Organizing Committee	
	• <i>Member</i> • <i>Co-Chair</i>	September 2017 - August 2018 September 2018 - August 2019
	Pitt Philosophy Mentor Program Organizing Committee	
	• <i>Co-Chair</i> • <i>Chair</i>	September 2017 - August 2018 September 2018 - August 2019
LANGUAGE	Mandarin, English, Classical Chinese, Python, R, Java, Scala, L ^A T _E X	
OUTREACH	Causation, Causal Direction and Causal Discovery	
	• <i>Three Guest Lectures at University of Chinese Academy of Sciences, Beijing, China</i>	May 2026
	Philosophy Unplugged	
	• <i>A series of animated videos about philosophy in Chinese for general audience</i> • <i>Access through BiliBili or YouTube</i>	Sep 2020 - March 2021
	Induction and Causation in Science	
	• <i>Two Guest Lectures for Undergraduate Students at Central University of Finance and Economics, Beijing, China</i>	May 2021
	Science and Society	
	• <i>Five Lectures for Advanced High School Students at Veritas China</i>	July - August 2020
	Equality in Contemporary Political Philosophy: Utilitarianism and Rawls	
	• <i>Online Public Lecture</i>	March 2020
GRADUATE COURSEWORK	Logic and AI	
	Proof Theory	Jeremy Avigad (CMU), Fall 2017
	Computability and Definability †	Jeremy Avigad (CMU), Spring 2018
† INDICATES AUDIT	Causal Discovery, Statistics and Machine Learning	Peter Spirtes (CMU), Spring 2019
	Algorithm Design	Kirk Pruhs, Fall 2021
	Machine Learning	Joseph Yurko, Fall 2021
	Natural Language Processing	David Mortensen and & Kemal Oflazer (CMU), Fall 2021
	Deep Learning	Bhiksha Raj (CMU), Spring 2022
	Network Analysis	Morgan Frank, Fall 2022
	Philosophy of Science	
	General Philosophy of Science	John Norton, Fall 2017
	Foundations of Statistics	Teddy Seidenfeld (CMU), Spring 2018
	Formal Epistemology of Science †	Kevin Kelly (CMU), Spring 2018
	Perception	Mazviita Chirimuuta, Spring 2019
	Theoretical & Practical Philosophy	
	Truth and Paradox	Anil Gupta, Fall 2017
	Practical Reason	Jed Lewinsohn & Michael Thompson, Fall 2017

Metaphysics and Epistemology
Philosophy of Language
Political Philosophy

Dmitri Gallow, Spring 2018
Robert Brandom, Fall 2018
Japa Pallikkathayil, Fall 2019

History of Philosophy

Carnap and Quine
Philosophy in Late Antiquity: from Philo to Philoponus
Kant: *Critique of Pure Reason*
Hegel: *The Phenomenology of Spirit* †
Kant: *Critique of Practical Reason* †

Thomas Ricketts, Spring 2018
Christian Wildberg, Fall 2018
Stephen Engstrom, Fall 2018
John McDowell, Spring 2019
Stephen Engstrom, Fall 2019

Economics

Mathematical Methods for Economists 1
Mathematical Methods for Economists 2
Advanced Microeconomics 1: General Equilibrium
Advanced Microeconomics 2: Game Theory
Evolutionary Game Theory
Political Economics †
Financial Economics
Advanced Econometrics 2: Time Series
Social Choice Theory †
Networks and Complexity †

Tymofiy Mylovanov, Summer 2019
Sofia Moroni & Roe Teper, Fall 2019
Roe Teper & Luca Rigotti, Fall 2019
Richard van Weelden, Spring 2020
Kevin Zollman (CMU), Fall 2019
Richard van Weelden, Fall 2020
Bryan Routledge (CMU), Spring 2021
Yiru Wang, Spring 2023
Tomas Sjöström & Andrew Mackenzie (Rutgers), Spring 2026
Sopher Barry (Rutgers), Fall 2026

REFERENCES

Kevin Zollman (Dissertation Co-Chair)
Herbert A. Simon Professor of Philosophy
and Social and Decision Sciences
Carnegie Mellon University
kzollman@andrew.cmu.edu

Kevin Dorst (Dissertation Examiner)
Associate Professor of Philosophy
Massachusetts Institute of Technology
kmdorst@mit.edu

James Shaw (Dissertation Co-Chair)
Professor of Philosophy
University of Pittsburgh
James.A.R.Shaw@gmail.com

Stephen Engstrom (Teaching)
Distinguished Professor of Philosophy
University of Pittsburgh
engstrom@pitt.edu